

Lecture 19

Determinant (II)

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Review and Outline

3 mins

Today's Lecture: Outline

Main topic: Determinant

1. Desired Properties of Determinants by Geometry
2. Properties of Determinant

Strang's book: Sec 8.1, 8.2

After the lecture, you should be able to

1. Tell the geometrical meaning of determinant
2. Use properties of determinants to compute determinants

Advanced goal (optional):

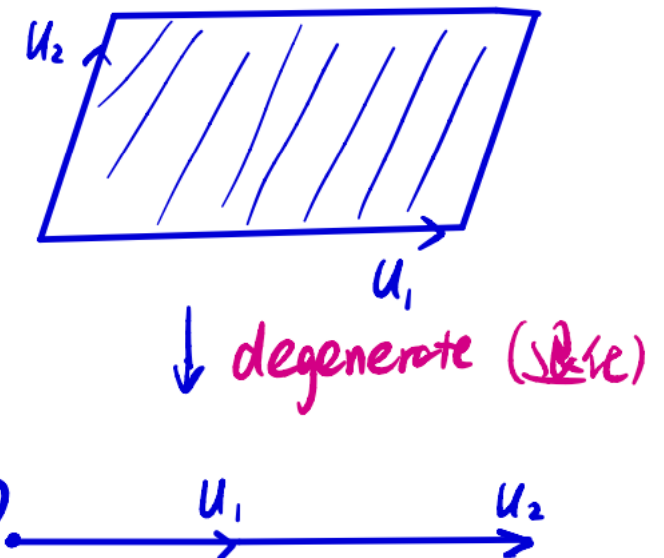
Relate algebraic properties to geometry.

Last Time: Geometry of Determinant

Question 18.2: WHAT is $ad - bc$?

Area of the parallelogram
(formed by two column vectors from the matrix)!

$ad - bc = 0$ means area = 0.



Question 18.1: [extension of invertibility condition]

Can we extend $(ad - bc)$ to $n \times n$ matrix, i.e. find function $\det()$ s.t.

$\det(A) \neq 0$ iff A is invertible?

$\det(A)$ = "Volume" of the polytope formed by $a_1, a_2, \dots, a_n \in \mathbb{R}^n$.

But how to compute the volume/determinant?

Part I Desired Properties of Determinant: Decomposition

22 mins

Extension to Higher Dimension

Question 18.1: [extension of invertibility condition]

Can we extend $(ad - bc)$ to $n \times n$ matrix, i.e. find function $\det()$ s.t.
 $\det(A) = 0$ iff A is invertible?

“Volume”!

How to compute the “volume”?

We will take a **special route**:

--**Properties**: We explore properties a “volume” should satisfy

--**Formula**: Based on the properties, we derive a method to compute “volume” (in geometry).

This method also defines “determinant” (in algebra).

Recall: How to Compute Area from Beginning?

→ How did we calculate area of  ?

- 1) Simplify: Relate  to 

- 2) Basic "area": Know areas of 

- } ✓

Determinant is Linear in Each Row

Notation: Denote $|A| = \det(A)$

Property P1:

The determinant is a linear function of each row separately.

When $n = 2$:

(P1.1) multiply row 1 by any number t
$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

(P1.2) Add row 1 of A to row 1 of B :
$$\begin{vmatrix} a_1 + a_2 & b_1 + b_2 \\ c & d \end{vmatrix} = \begin{vmatrix} a_1 & b_1 \\ c & d \end{vmatrix} + \begin{vmatrix} a_2 & b_2 \\ c & d \end{vmatrix}$$

General n :

(P1.1)
$$\begin{vmatrix} tv_1 \\ v_2 \\ \dots \\ v_m \end{vmatrix} = t \begin{vmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{vmatrix}$$

(P1.2)
$$\begin{vmatrix} v_1 + u_1 \\ v_2 \\ \dots \\ v_m \end{vmatrix} = \begin{vmatrix} v_1 \\ v_2 \\ \dots \\ v_m \end{vmatrix} + \begin{vmatrix} u_1 \\ v_2 \\ \dots \\ v_m \end{vmatrix}$$

We showed 1st row; same relations for other rows.

Steps to Understand the Formula

We do NOT fully prove the property.

To understand the property:

Step 1: Verify the property in the 2×2 case.

Step 2: Explain the property geometrically in the 2×2 case.

P1.1 Scaling a Row: Step 1

P1.1: Multiply one row of A by any number:

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \cdot \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Step 1: Verify algebraically:

$$\begin{aligned} \text{LHS} &= \begin{vmatrix} ta & tb \\ c & d \end{vmatrix} \\ &= ta \cdot d - tb \cdot c \\ &= t(ad - bc) \end{aligned}$$

$$\begin{aligned} \text{RHS} &= t \cdot \begin{vmatrix} a & b \\ c & d \end{vmatrix} \\ &= t(ad - bc) \end{aligned}$$

Step 2: Geometry: for $n=2$ (P1.1) Scaling a Row

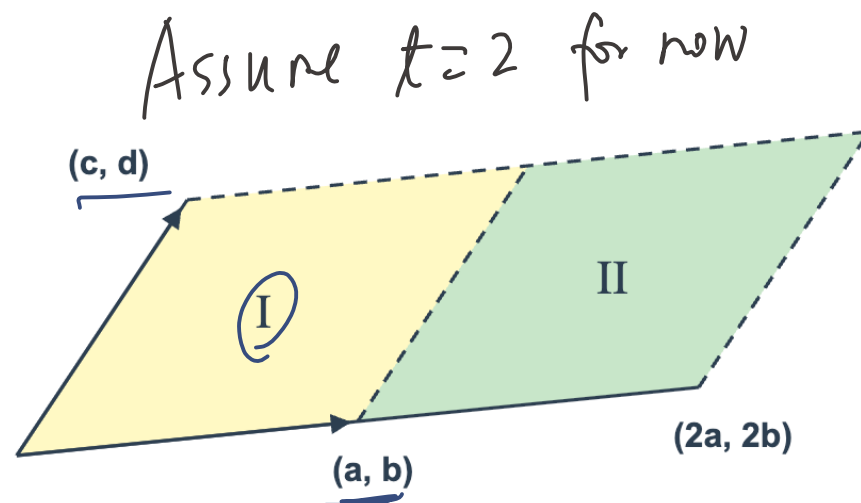
Step 2 for P1.1:

Explain the property geometrically in the 2×2 case

(P1.1) **multiply row 1 by any number t**
$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Denote $\text{Para}[(a,b),(c,d)]$ as the **parallelogram formed by the two vectors** (a,b) and (c,d) .

$$\begin{vmatrix} 2a & 2b \\ c & d \end{vmatrix} \stackrel{?}{=} 2 \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$



$$\begin{aligned} \text{LHS} &= \text{Area}(\text{Para}[(2a, 2b), (c, d)]) \\ &= \text{Area of I} + \text{Area of II} = 2 \cdot (\text{Area of I}) \\ &= 2 \text{Area}(\text{Para}[(a, b), (c, d)]) \\ &= 2 \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \text{RHS}. \end{aligned}$$

multiply one side by 2,
 $\Rightarrow$ area becomes 2 times the
original one

日期: /

$3\vec{a}$

$\vec{a}$

Polytope

multiply one side by 3,

new volume will be

3 times the original
volume

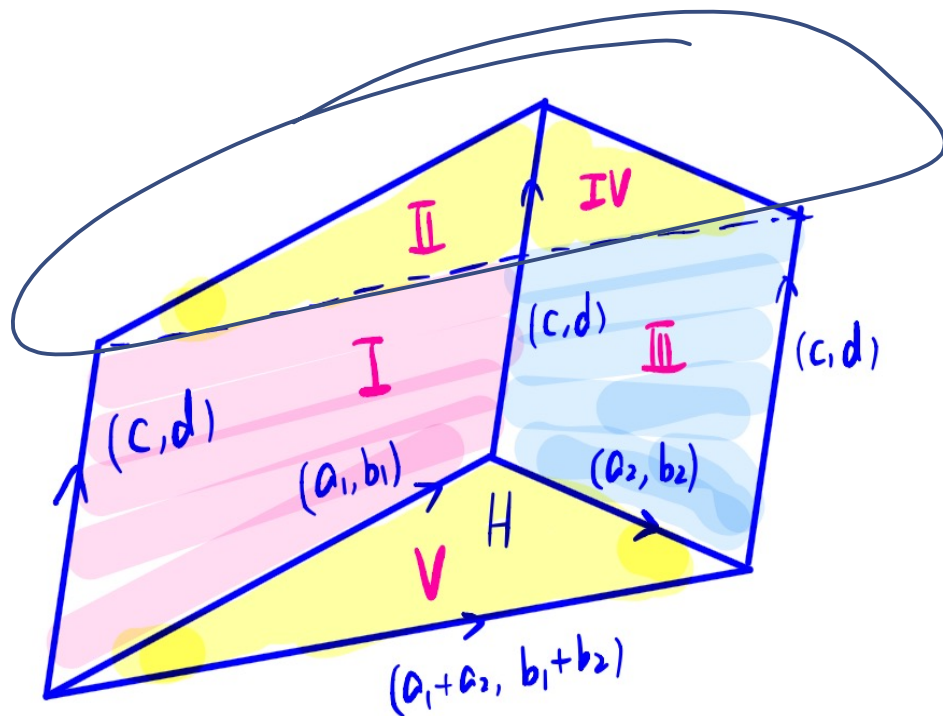
P1.2 Summing a Row

P1.2: Add row 1 of A to row 1 of B : $\begin{vmatrix} a_1 + a_2 & b_1 + b_2 \\ c & d \end{vmatrix} = \begin{vmatrix} a_1 & b_1 \\ c & d \end{vmatrix} + \begin{vmatrix} a_2 & b_2 \\ c & d \end{vmatrix}$

Step 1: Algebra proof for $n=2$.

$$\begin{aligned} \text{LHS} &= (a_1 + a_2)d - (b_1 + b_2)c \\ &= a_1d + a_2d - b_1c - b_2c \\ &= (a_1d - b_1c) + (a_2d - b_2c) = \text{RHS} \end{aligned}$$

Step 2: Geometry proof for $n=2$.



$$\begin{aligned} \text{LHS} &= \text{I} + \text{III} + \text{V} \\ \text{RHS} &= (\text{I} + \text{II}) + (\text{III} + \text{IV}) \\ &= \text{I} + \text{III} + (\text{II} + \text{IV}) \end{aligned}$$

Congruent triangulars (全等三角形) have the same area:

$$\text{V} = \text{II} + \text{IV}$$

Thus $\text{LHS} = \text{RHS}$.

Exercise: Multiply Both Rows

Example 1 (2×2 case):

$$\det(2A) \neq 2 \det(A)$$

$$\begin{vmatrix} 2a & 2b \\ 2c & 2d \end{vmatrix} = 2 \begin{vmatrix} a & b \\ 2c & 2d \end{vmatrix} = 2 \times 2 \begin{vmatrix} a & b \\ c & d \end{vmatrix} = 4 \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Example 2 (3×3 case):

$$\begin{vmatrix} 3\vec{v}_1 \\ 3\vec{v}_2 \\ 3\vec{v}_3 \end{vmatrix} = 3 \begin{vmatrix} \vec{v}_1 \\ 3\vec{v}_2 \\ 3\vec{v}_3 \end{vmatrix} = 3 \times 3 \begin{vmatrix} \vec{v}_1 \\ \vec{v}_2 \\ 3\vec{v}_3 \end{vmatrix} = 27 \begin{vmatrix} \vec{v}_1 \\ \vec{v}_2 \\ \vec{v}_3 \end{vmatrix}$$

Exercise: Sum instead of Single-row-sum

$$\det(A+B) = \det(A) + \det(B)?$$

$$\begin{vmatrix} a_1 + a_2 & b_1 + b_2 \\ c_1 + c_2 & d_1 + d_2 \end{vmatrix} \stackrel{?}{=} \begin{vmatrix} a_1 & b_1 \\ c_1 & d_1 \end{vmatrix} + \begin{vmatrix} a_2 & b_2 \\ c_2 & d_2 \end{vmatrix}$$

$$\text{LHS} = (a_1 + a_2)(d_1 + d_2) - (b_1 + b_2)(c_1 + c_2)$$

4 4
8 term

$$\text{RHS} = (a_1 d_1 - b_1 c_1) + (a_2 d_2 - b_2 c_2).$$

4 terms

Hint: compare # of terms on both sides.

Decompose with linearity on the first row

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} =$$

Application of P1 for Computing Determinant

For computing determinant, we only need a special case of P1:
represent each row by LC of the standard basis of $\mathbb{R}^3$.

$$\underbrace{[a_1 \quad a_2 \quad a_3]}_{(=a_1 \vec{e}_1)} = \underbrace{[a_1, 0, 0]}_{+ a_2 \vec{e}_2} + \underbrace{[0, a_2, 0]}_{+ a_3 \vec{e}_3} + \underbrace{[0, 0, a_3]}$$

Then

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} \underbrace{a_1} & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & \underbrace{a_2} & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & 0 & \underbrace{a_3} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

To process each term, we need two more properties.

Part II Desired Properties of Determinant: Size Reduction and Others

Calculation of Decomposition (1st Appearance)

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & a_2 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

Note that

$$\begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} = \begin{bmatrix} a_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ b_1 \\ c_1 \end{bmatrix}$$

To handle 1st Term: Decompose via 1st column:

$$\begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ 0 & b_2 & b_3 \\ 0 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

*block diagonal
matrix*

*matrix
with zero-row*

Part II.1 Block Diagonal Determinant and Zero-Row Determinant

10 mins

Handling Zero-row Term

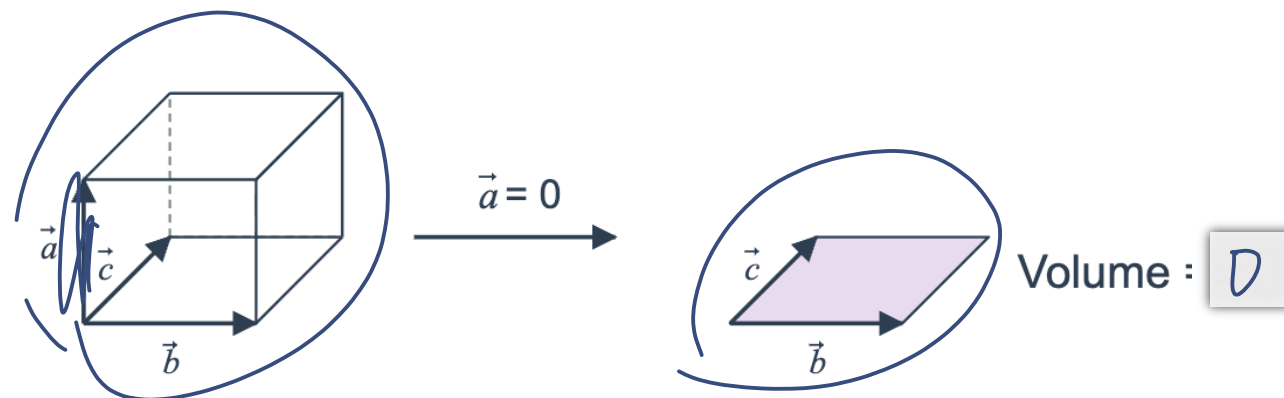
Corollary of P1:

If one row or column of a square matrix A is zero, then $\det(A) = 0$.

e.g. $\begin{vmatrix} 0 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0.$

Geometry intuition:

In 3D, the polytope



Algebraic proof:

Denote $\mathbf{p} = \begin{vmatrix} 0 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$

Note, **vector 0** = 2×0 , thus by P1, $\mathbf{p} = \begin{vmatrix} 0 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 2 \begin{vmatrix} 0 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 2\mathbf{p}.$

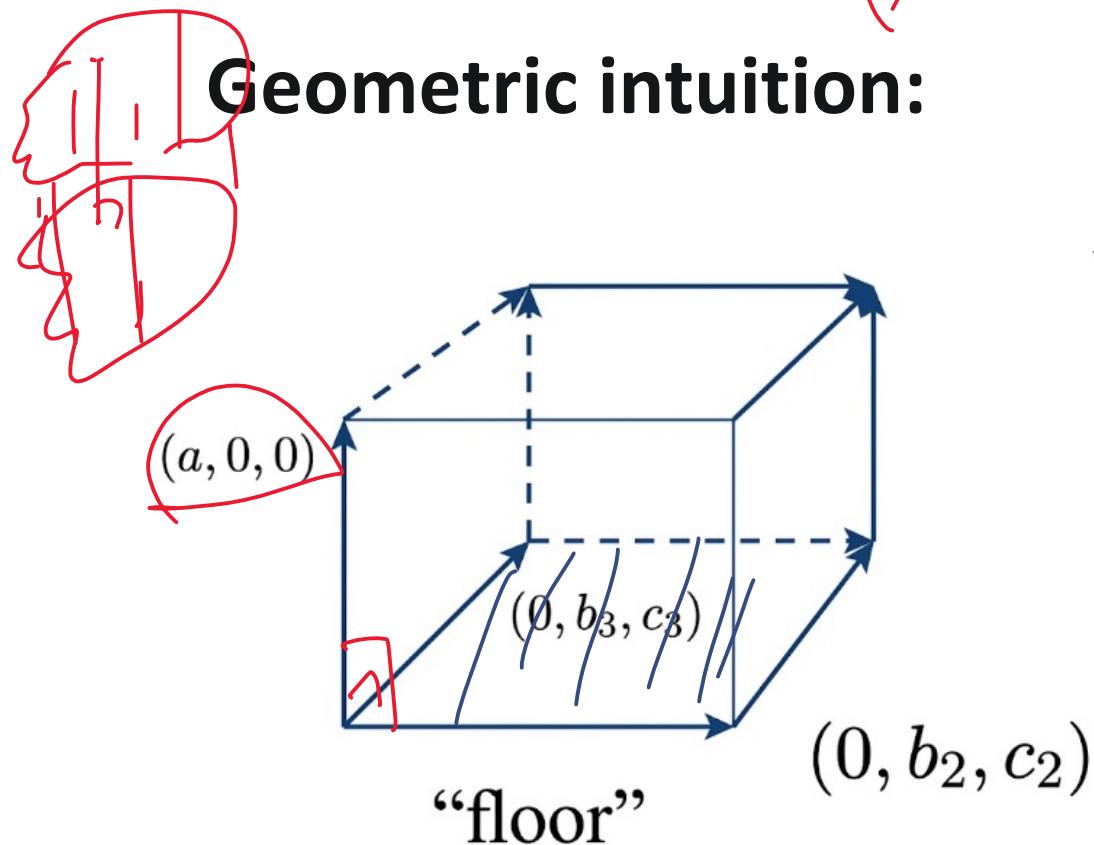
Thus $\mathbf{p} = 0$.

Handling Block Diagonal Matrix

P2 If an $n \times n$ matrix $A = \begin{bmatrix} h & \mathbf{0}_{1 \times (n-1)} \\ \mathbf{0}_{(n-1) \times 1} & B \end{bmatrix}$, then $\det(A) = h \cdot \det(B)$.

We verify it for $n=3$, i.e., $\begin{vmatrix} a & 0 & 0 \\ 0 & b_2 & b_3 \\ 0 & c_2 & c_3 \end{vmatrix} = a \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}$

Geometric intuition:



$$\begin{aligned} & \text{Volume of this polytope} \\ &= \text{height} \times \text{area of para}[(0, b_2, c_2), (0, b_3, c_3)] \\ &= a \times \text{area of para}[(b_2, c_2), (b_3, c_3)] \\ &= a \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} \end{aligned}$$

Calculation of Decomposition (2nd Appearance)

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & a_2 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

To handle 1st Term: Decompose via 1st column:

$$\begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ 0 & b_2 & b_3 \\ 0 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}.$$

*block diagonal
term* *Zero-row
term*

*Reducing 3*3
to 2*2!*

Calculation of Decomposition (3rd Appearance)

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & a_2 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

1ST term: $\begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}.$ Reducing 3*3 to 2*2!

2nd term: $\begin{vmatrix} 0 & a_2 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \stackrel{\text{swap}}{=} \begin{vmatrix} a_2 & 0 & 0 \\ b_2 & b_1 & b_3 \\ c_2 & c_1 & c_3 \end{vmatrix} = a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}$

Remaining question:

how does swapping column change determinant?

Part II.2 Swapping Rows/Columns

15 mins

General Dimension: Swapping Rows/Columns

Property P3: [row/column exchange]

Swapping two rows (or two columns) changes the sign of the determinant.

Algebraic Verification for 2 by 2 case:

$$\begin{vmatrix} x & y \\ z & w \end{vmatrix} = xw - yz,$$

$$\begin{vmatrix} y & x \\ w & z \end{vmatrix} = yz - xw = - \begin{vmatrix} x & y \\ z & w \end{vmatrix}.$$

We will provide some geometric intuition next.

e.g. of P3: $\det([v_1, v_2]) = -\det([v_2, v_1])$.

$$\det([v_1, v_2, v_3]) = -\det([v_2, v_1, v_3])$$

$$\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$$

$$\begin{aligned} \det([v_1, v_2, v_3]) &= -\det([v_3, v_2, v_1]) \\ &= -(-\det([v_3, v_1, v_2])) = \det([v_3, v_1, v_2]). \end{aligned}$$

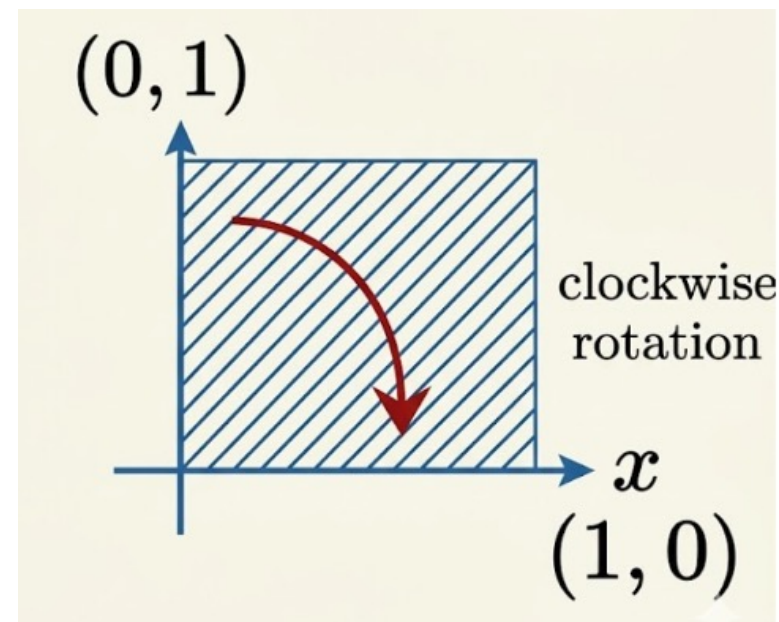
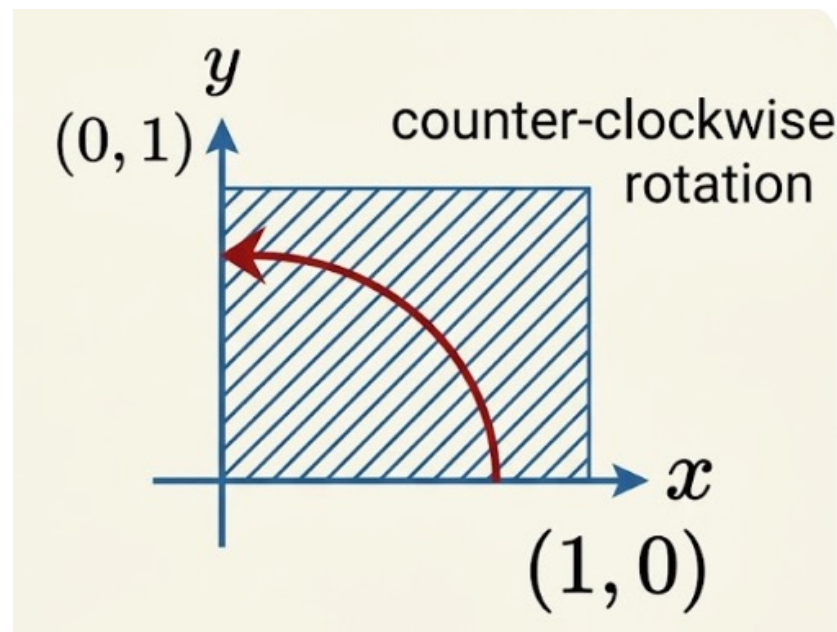
Geometry Interpretation of Swapping Columns

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

$$\det([e_1, e_2]) = 1.$$

$$\det([e_2, e_1]) = -1.$$

Two determinants are different; Both are rectangulars formed by e_1, e_2 .
How to interpret the difference?

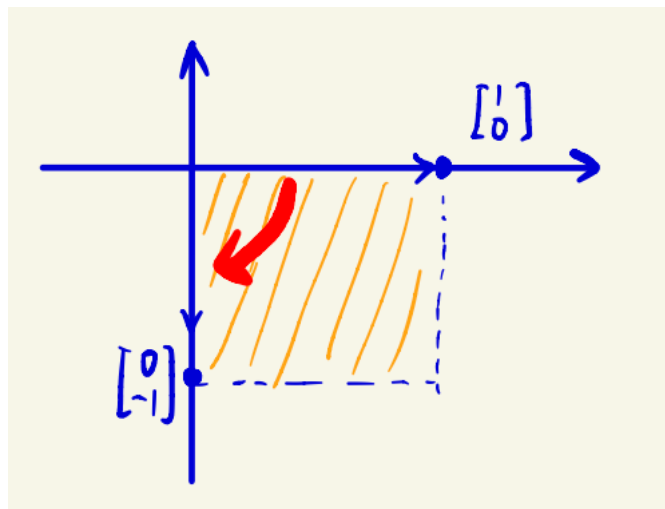


New interpretation: Oriented Area (有向面积)

Verification by Another Example

What is $\det([e_1, -e_2])$?

Geometric perspective: obtain -1.



$e_1 \rightarrow -e_2$
clockwise.

Under the new interpretation: **oriented area = -1**

Algebraic perspective: obtain -1.

$$\det([e_1, -e_2]) = \begin{vmatrix} 1 & 0 \\ 0 & -1 \end{vmatrix} = -1.$$

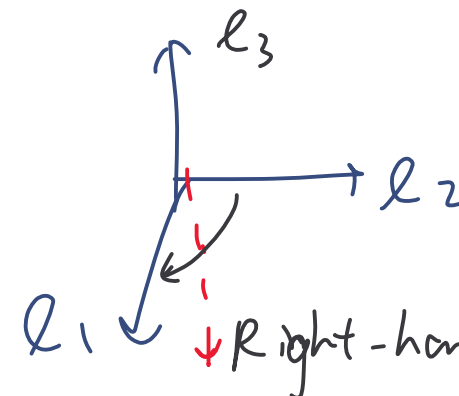
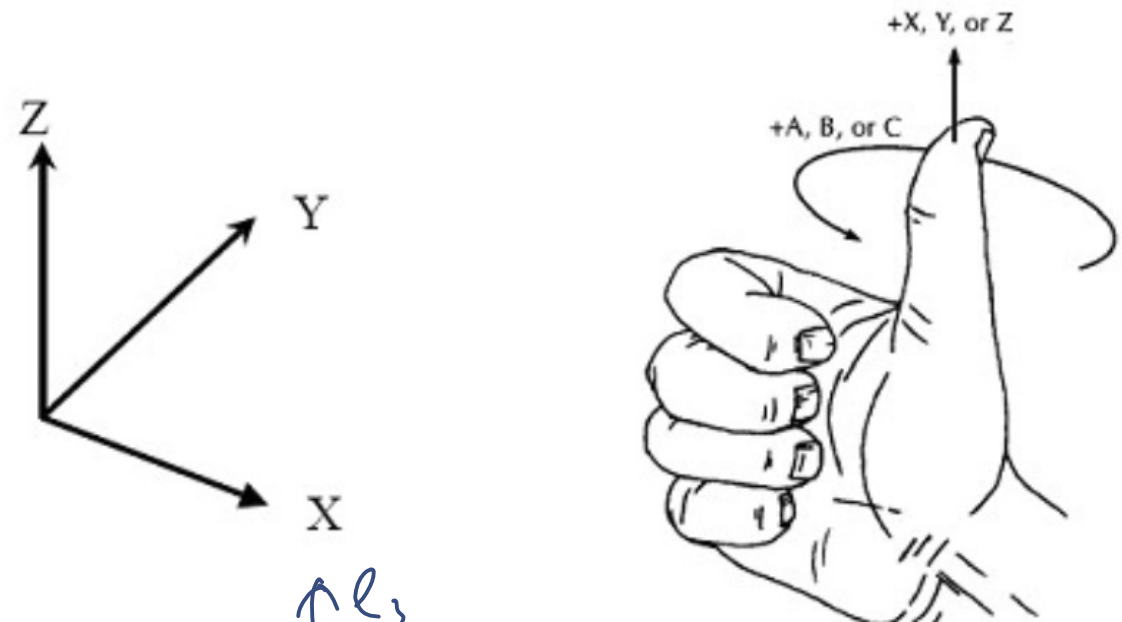
3D Case: Handedness

*3D case:
Right handedness.*

$$\det(e_1, e_2, e_3) = 1.$$

$$\det(e_2, e_1, e_3) = -1.$$

Swapping two vectors leads to flipped sign.



Calculation of Decomposition (4th Appearance)

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & a_2 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

$$1^{st} \text{ term: } \begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}.$$

*Reducing 3*3 to 2*2!*

$$2^{nd} \text{ term: } \begin{vmatrix} 0 & a_2 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = - \begin{vmatrix} a_2 & 0 & 0 \\ b_2 & b_1 & b_3 \\ c_2 & c_1 & c_3 \end{vmatrix} = -a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}$$

Calculation of Decomposition (5th Appearance)

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & a_2 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

1ST term: $\begin{vmatrix} a_1 & 0 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix}.$ Reducing 3*3 to 2*2!

2nd term: $\begin{vmatrix} 0 & a_2 & 0 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = - \begin{vmatrix} a_2 & 0 & 0 \\ b_2 & b_1 & b_3 \\ c_2 & c_1 & c_3 \end{vmatrix} = -a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix}$

3rd term: $\begin{vmatrix} 0 & 0 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = -1 \begin{vmatrix} 0 & a_3 & 0 \\ b_1 & b_3 & b_2 \\ c_1 & c_3 & c_2 \end{vmatrix} = -(-1) \begin{vmatrix} a_3 & 0 & 0 \\ b_3 & b_1 & b_2 \\ c_3 & c_1 & c_2 \end{vmatrix} = a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$

We get the expression of the determinant of 3×3 matrix:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}.$$

Prepare to generalize: Notation of Cofactor

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Define $M_{11} = \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} =$ matrix obtained by deleting
1st row & 1st column.

$$M_{12} = \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} =$$

$$M_{13} = \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} =$$

$$\begin{aligned} \det(A) &= a_{11}|M_{11}| - a_{12}|M_{12}| + a_{13}|M_{13}| \\ &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}). \end{aligned}$$

Part III Formal Introduction of Determinant

15 mins

Part III.1 Definition and Basic Properties of Determinant

9 mins

Definition of $\det(A)$

Let $A \in \mathbb{R}^{n \times n}$ be a real square matrix

Denote by $M_{ij} \in \mathbb{R}^{(n-1) \times (n-1)}$ a matrix formed by deleting the i -th row and j -th column of A , called the **minor (余子式)** of a_{ij} .

Definition 19.1 (Determinant)

For a scalar $\alpha \in \mathbb{R}$, define $\det(\alpha) = \alpha$.

For any $A \in \mathbb{R}^{n \times n}$ with $n \geq 2$ define $|A| = \det(A)$ as

$$\det(A) = \sum_{j=1}^n (-1)^{1+j} \det(M_{1j}) a_{1j} = a_{11} \det(M_{11}) - a_{12} \det(M_{12}) + a_{13} \det(M_{13}) - \dots + (-1)^{n+1} a_{1n} \det(M_{1n})$$

$\downarrow$ $(n-1) \times (n-1)$ $\downarrow$ $(n-2) \times (n-2)$

This is a **recursive definition**! [递归方式的定义]

$$|M_{11}| = a_{22} |M'_{22}| - a_{23} |M'_{23}| + \dots$$

$\downarrow$ $(n-1) \times (n-1)$ $\downarrow$ $(n-2) \times (n-2)$

Examples of Determinant

Example

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \underline{ad} - \underline{bc} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{aligned} |A| &= \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = \underline{a} \begin{vmatrix} e & f \\ h & i \end{vmatrix} - \underline{b} \begin{vmatrix} d & f \\ g & i \end{vmatrix} + \underline{c} \begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ &= \underline{aei} + \underline{bfg} + \underline{cdh} - ceg - bdi - afh \end{aligned}$$

Property: Expand along any row or column

Proposition (Laplace Expansion)

Let A be an $n \times n$ matrix with entries a_{ij} . For any fixed row k (where $1 \leq k \leq n$) or any fixed column j (where $1 \leq j \leq n$), the determinant $\det(A)$ can be expressed by "expanding" along that row or column as follows:

Column j expansion: $\det(A) = \sum_{k=1}^n (-1)^{k+j} \det(M_{kj}) a_{kj},$

Row k expansion: $\det(A) = \sum_{j=1}^n (-1)^{k+j} \det(M_{kj}) a_{kj}.$

Example 1 (expansion along the 2nd row, $k=2$):

$$\det(A) = \underbrace{-}_{(-1)^{2+1}} a_{2,1} |M_{2,1}| + a_{2,2} |M_{2,2}| - a_{2,3} |M_{2,3}| + \cdots + (-1)^{n+2} a_{2,n} |M_{2,n}|$$

Example 2 (expansion along the 3rd column, $j=3$):

$$\det(A) = (-1)^{1+3} a_{1,3} |M_{1,3}| + (-1)^{2+3} a_{2,3} |M_{2,3}| + \cdots + (-1)^{n+3} a_{n,3} |M_{n,3}|$$

Properties of Determinant

Property P19.0 [transpose] $\det(A^T) = \det(A)$

Skip the proof.

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Property P19.1 [row-linear and column-linear]

The determinant is a linear function of each row.

The determinant is a linear function of each column.

Verify with Laplace expansion formula).

Property P19.2 [reduction]

If an $n \times n$ matrix $A = \left[\begin{array}{c|c} h & \mathbf{0}_{1 \times (n-1)} \\ \hline \mathbf{0}_{(n-1) \times 1} & B \end{array} \right]$, then $\det(A) = h \cdot \det(B)$.

Just by definition.

Example: Computing via the Laplace Expansion

Example

Evaluate

$$\begin{vmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{vmatrix}$$

Trick: pick rows/columns with many 0's.

$$(-1)^{5} 2 \begin{vmatrix} 2 & 3 & 0 \\ 4 & 5 & 0 \\ 1 & 0 & 3 \end{vmatrix} = -2 \cdot (-1)^{6} 3 \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = -2 \cdot 3 \cdot (10 - 12) = 12$$

$(-1)^{4+1}$ since
2 is (4,1)-th entry

$(-1)^{3+3}$ since
3 is the (3,3)-th entry

Part III.2 Determinant and Elementary Row Operations

6 mins

Three Elementary Row Operations

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

det $\times$
1 or -1?

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{\alpha R_2} \begin{bmatrix} a & b \\ \alpha c & \alpha d \end{bmatrix}$$

det $\times$
1 or α ?

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \xrightarrow{\beta R_1 + R_2} \begin{bmatrix} a & b \\ c + \beta a & d + \beta b \end{bmatrix}$$

det $\times$

$$= a(d + \beta b) - b(c + \beta a)$$

$$= ad + \cancel{\beta ab} - bc - \cancel{\beta ab}$$

$$= ad - bc.$$

1, β or $\beta + 1$?

Analysis of “Adding Scaled Row” Using Properties

What about > 2 -dim? We consider 3 by 3.

We consider “adding scaled column”; for rows, similar.

Consider a matrix $A = [a_1, a_2, a_3]$ with column vectors a_1, a_2, a_3 .

change to $B = [a_1, a_2, a_3 + \beta a_1]$.

$$\begin{aligned}\det(B) &= \det(a_1, a_2, a_3 + \beta a_1) \\ &= \det(a_1, a_2, a_3) + \beta \det(a_1, a_2, a_1),\end{aligned}$$

[row linear]

$$\det(a_1, a_2, a_1) = -\det(a_1, a_2, a_1) \implies \det(a_1, a_2, a_1) = 0$$

[swapping columns changes sign]

Thus $\det(B) = \det(a_1, a_2, a_3) = \det(A)$.

Determinant through elementary operations

Property P19.3 [interchange] $\det(E_{R_i \leftrightarrow R_j} A) = -\textit{det}(A)$

Corollary: [Addition] $\det(E_{\beta R_j + R_i} A) = \textit{det}(A)$

“Add a Scaled Row to Another” does not change the determinant.

Corollary [Scaling] $\det(E_{\alpha R_i} A) = \alpha \cdot \textit{det}(A)$

Example

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 4 \end{bmatrix}.$$

$$\det(A) = \det \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix} \quad (R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - R_1)$$

$$= \det \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \quad (R_3 \leftarrow R_3 - R_2)$$

$$= 2 \det \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad (\underline{R_3} \leftarrow \frac{1}{2} R_3)$$

$$= 2 \det \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{backward: } R_1 \leftarrow R_1 - R_3, R_2 \leftarrow R_2 - R_3)$$

$$= 2 \det \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{backward: } R_1 \leftarrow R_1 - 2R_2)$$

[P19.2] $= 2 \cdot 1 = 2$



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:

Summary Today (Instructor)

One sentence summary:

We learned definition and derivation of determinants.

Detailed summary:

Computing “volume”:

—Overall idea: “Simplification” framework

—Properties:

linear over rows/columns

swapping columns changes sign.

Definition: Laplacian expansion over 1st row.

Basic properties: $\det(A^T) = \det(A)$;

determinant after row operations

Appendix: Oriented Volume

Orientation in High Dimension

I'm teaching a freshman class. I argued that determinant is the volume of the parallelotope. But there is an issue: the volume is always positive. How come the determinant is negative? So I used the figure to illustrate the 2D case. What about high dimension?

Below is a **freshman-level** explanation of orientation in higher dimensions that does **not** rely on a physical "hand rule." Instead, it uses the **standard basis** as the reference for what we call "positive orientation." Any other basis' orientation is determined by *how many times* we swap vectors to get from the standard basis to that new basis.

1. The Standard Basis as Our Reference

In n -dimensional space $\mathbb{R}^n$, we have the standard basis:

$$\mathbf{e}_1 = (1, 0, 0, \dots, 0), \quad \mathbf{e}_2 = (0, 1, 0, \dots, 0), \quad \dots, \quad \mathbf{e}_n = (0, 0, \dots, 0, 1).$$

By **definition**, we declare that $(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ is **positively oriented** or "right-handed." This is just a convention—a choice—so we have a baseline.

Orientation in High Dimension

2. How to Tell if Another Ordered Basis Is “Flipped”

Given any other ordered basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$, we can always convert the standard basis into $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ through elementary steps:

1. **Swapping** pairs of basis vectors (think: swap $\mathbf{e}_i$ and $\mathbf{e}_j$).
2. **Scaling** a basis vector by a nonzero scalar.
3. **Adding** one basis vector to another.

Out of all these steps, **only swapping** two vectors changes the “orientation.”

- Every swap is like flipping a mirror image in at least one direction (in 2D or 3D, that is more visually obvious, but we treat it algebraically in higher dimensions).
- Each swap flips the sign of the determinant.

Hence:

- If we can reach $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ from $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ using an **even** number of swaps (and any number of scalings/additions), then $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is also **positively oriented**.
- If it requires an **odd** number of swaps, then $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is **negatively oriented**.

This is exactly why **swapping any two columns** of a matrix flips the sign of the determinant: it is that single “odd swap.”

Key point: We do *not* need to physically see a “hand rule” in 4D or 10D; the sign of the determinant is the purely algebraic record of whether we used an even or odd number of swaps to go from the standard basis to our new basis.

Orientation in High Dimension

3. Sign of the Determinant: Oriented Volume

1. **Absolute value** of $\det(A)$ tells you how much the transformation “stretches” or “shrinks” volumes.
2. The **sign** of $\det(A)$ tracks whether the transformation can be done with an *even* or *odd* number of swaps from $(\mathbf{e}_1, \dots, \mathbf{e}_n)$ to the new set of axes.
 - **Positive** $\rightarrow$ an even number of swaps (no net flip relative to the standard basis).
 - **Negative** $\rightarrow$ an odd number of swaps (a flip relative to the standard basis).

Thus, for a freshman class, you can say:

“We picked the standard basis to be *the* positively oriented frame. If your matrix’s column vectors can be rearranged from the standard basis using an odd number of swaps, that’s one *net flip*, so the determinant is negative. If it’s an even number of swaps, it’s not flipped overall, so the determinant is positive.”

No complicated “hand rule” is required—you simply keep track of **swap count** from a declared reference. That’s how we know whether we ‘have flipped orientation or have not.’